COIMBATORE INSTITUTE OF TECHNOLOGY

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19MAME23 – LARGE LANGUAGE MODELS

ASSIGNMENT – III

CASE STUDY

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Corporate Ticketing System & Health Issue Classification

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Executive Summary

This case study examines the application of Large Language Models (LLMs) to solve the challenge of identifying and routing health-related tickets in a corporate ticketing system. With the advent of medical counsel and hospital partnerships, organizations must now automatically classify tickets without historical labeled data. This document presents an analysis of the text classification problem, challenges with unlabeled data, the role of LLMs, and a proposed solution design using zero-shot and few-shot learning with prompt engineering.

1. Text Classification Problem Identification

1.1 Why This is a Text Classification Problem

Identifying health-related tickets is fundamentally a **text classification problem** in NLP for the following reasons:

a) Unstructured Textual Input: Employee tickets are submitted as free-form text descriptions that must be analyzed to extract meaning and intent.

b) Categorical Output Requirement: The system must assign each ticket to predefined categories (health-related vs. non-health-related), which is the fundamental definition of classification.

c) Semantic Understanding Required: The classification task requires understanding context to differentiate between medical symptoms, technical issues, and administrative requests.

d) Natural Language Variability: The system must handle synonyms, medical terminology vs. layman's terms, and implicit health mentions.

e) Multi-Class Decision Making: Beyond binary classification, the system may route tickets to sub-categories like emergency medical, mental health, or injury reports.

1.2 Input, Output, and Classes

INPUT:

- Unstructured text (free-form employee ticket descriptions)

- Natural language strings (typically 10-500 words)
- Channels: Email, web portal, mobile app, chat systems
- Content: Subject line + detailed description

OUTPUT:

- Categorical classification label (HEALTH-RELATED / NOT-HEALTH-RELATED)
- Optional: Confidence score (0.0 to 1.0)
- Optional: Sub-category assignment

CLASSES:

Class	Description	Examples
Health-Related	Tickets requiring medical attention	Symptoms, injuries, mental health, insurance queries
Non-Health-Related	All other organizational tickets	IT issues, HR queries, facilities requests

Health-Related Sub-Categories: Emergency Medical, Routine Medical, Mental Health, Injury/Accident, Insurance/Benefits, Preventive Care

Non-Health-Related Sub-Categories: IT Support, HR Queries, Facilities, Finance

2. Challenges with Unlabeled Data

2.1 Challenges with Building Classification Systems Without Labeled Data

a) Inability to Train Supervised Models: Traditional ML algorithms require labeled examples. Without labels, there are no training examples, no decision boundaries, and no reference for "correct" behavior.

b) Cold Start Problem: Historical tickets exist but lack health-related labels. Manual labeling is expensive ($0.50-$2.00 per ticket), time-consuming, requires medical expertise, and risks inconsistent labeling.

c) Class Imbalance Uncertainty: Without labels, we cannot determine what percentage of tickets are health-related or if the distribution is highly imbalanced.

d) Feature Engineering Difficulty: Traditional ML requires manual feature extraction and domain expertise, with risk of missing implicit indicators.

e) Validation Impossibility: Cannot calculate accuracy metrics, perform cross-validation, or measure model improvement without labeled data.

f) Concept Drift: Health-related language evolves and varies by organizational culture and regional differences.

2.2 Why Traditional Supervised Learning Methods Are Difficult to Apply

1. Data Dependency

Model Type	Training Data Required
Logistic Regression	1,000-10,000 examples
Support Vector Machines	500-5,000 examples
Deep Neural Networks	10,000-100,000+ examples
BERT Fine-tuning	500-5,000 examples

2. Static Model Limitations: Cannot adapt to new terminology without retraining; lack reasoning capability for complex scenarios.

3. Context Understanding Deficiency: Traditional models struggle with semantic similarity, negation handling, and context-dependent meanings.

4. Annotation Cost and Time

Task	Estimated Cost	Time Required
Annotate 5,000 tickets	$2,500-$10,000	2-4 weeks
Medical expert review	$5,000-$15,000	3-6 weeks
Quality assurance	$1,000-$3,000	1-2 weeks
Total	**$10,500-$33,000**	**7-14 weeks**

5. Lack of Transfer Learning: External datasets face domain mismatch with organizational-specific terminology and ticket writing styles.

3. Role of Large Language Models

3.1 How LLMs Classify Health-Related Tickets Without Fine-Tuning

Large Language Models like GPT-4, Claude, and Gemini revolutionize this problem through their zero-shot and few-shot capabilities:

A. Pre-Trained World Knowledge

LLMs are trained on massive text corpora including medical literature, health websites, and conversational data. The model inherently understands medical terminology, symptoms, and context clues indicating health concerns.

B. Zero-Shot Classification

Zero-shot learning means the model classifies text without explicit training examples:

1. Provide task description in natural language
2. Define what “health-related” means

3. Give the ticket text to analyze
4. Model outputs classification based on pre-trained knowledge

Advantages: No training data required, fast deployment, flexible definitions, cost-effective.

C. Few-Shot Learning Enhancement

Few-shot learning improves accuracy by providing 3-10 examples in the prompt:

1. Include labeled examples in the prompt
2. Model identifies patterns from examples
3. Classifies new tickets based on learned patterns

Advantages: 10-20% accuracy improvement, better domain adaptation, handles edge cases.

D. Contextual Understanding

LLMs excel at:

- **Semantic Comprehension:** Synonyms ("ill" = "sick"), implicit references, medical jargon
- **Pragmatic Reasoning:** Literal vs. figurative ("sick of this meeting" vs. "feeling sick"), negation handling

3.2 Role of Prompt Engineering

Prompt engineering designs effective input instructions to guide LLM behavior.

Components of Effective Prompts:

Component	Purpose
Task Definition	State the classification objective clearly
Class Descriptions	Provide explicit definitions with boundaries
Output Format	Define expected output structure
Examples	Include diverse samples for few-shot learning
Edge Case Guidance	Address ambiguous scenarios

Advanced Techniques:

- **Chain-of-Thought (CoT):** Step-by-step reasoning
- **Role-Based Prompting:** Assign model a specific persona
- **Confidence Scoring:** Request probability estimates

Prompt Optimization Strategies:

Strategy	Purpose

A/B Testing	Compare prompt formulations
Iterative Refinement	Improve based on errors
Domain Contextualization	Add organization-specific context
Temperature Control	Lower temperature (0.1-0.3) for consistency

3.3 Example Prompt Structure

A production-ready prompt includes:

1. **System Role:** Define the AI as a ticket classification system
2. **Classification Task:** Binary health vs. non-health decision
3. **Health-Related Definition:** Physical symptoms, mental health, injuries, insurance queries, preventive care
4. **Non-Health Definition:** IT support, HR admin, facilities, access permissions
5. **Ambiguous Cases:** Rules for mixed-category tickets
6. **Output Format:** Classification label, confidence score, reasoning

Sample Classifications:

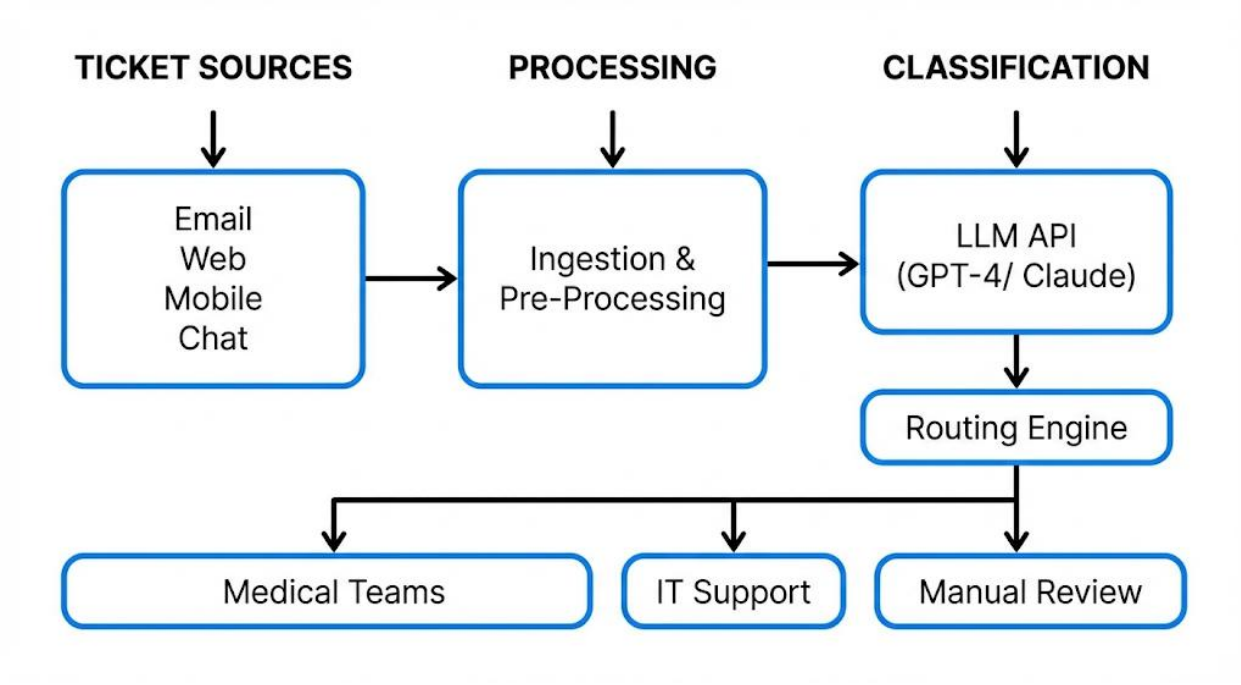
Ticket	Classification	Reason
“Headache after 2 hours of work”	HEALTH	Physical symptom
“Reset password for HR portal”	NON-HEALTH	IT access issue
“Feeling anxious before presentations”	HEALTH	Mental health concern
“Conference projector not working”	NON-HEALTH	Facilities issue
“Sick of long meetings”	NON-HEALTH	Figurative expression

4. Proposed Solution Design

4.1 High-Level Architecture

The proposed solution uses a **cloud-based LLM API architecture** with intelligent routing:

System Flow Diagram:



System Components:

1. **Ticket Ingestion Layer:** API, validation, duplicate detection, metadata extraction
2. **Pre-Processing Layer:** Text cleaning, normalization, PII redaction
3. **LLM Classification Engine:** Zero/few-shot classification, confidence scoring
4. **Routing Decision Engine:** Rule-based routing based on classification
5. **Assignment & Notification:** Team assignment, alerts, status updates
6. **Feedback Loop:** Agent feedback, prompt refinement, metrics dashboard

4.2 Data Flow Summary

Step	Action	Output
1. Submission	Employee submits ticket	Ticket ID + metadata
2. Pre-processing	Clean text, validate, redact PII	Normalized text
3. Classification	LLM API call with prompt	Classification + confidence
4. Routing	Apply decision rules	Destination team + priority
5. Assignment	Assign to agent, notify	Updated ticket status
6. Resolution	Agent handles, provides feedback	Classification accuracy log

4.3 Medical Ticket Detection Mechanism

Multi-Layered Detection Approach:

Layer 1 - LLM Classification: Primary zero-shot/few-shot classification with confidence scoring

Layer 2 - Keyword Boosting: For medium-confidence results (0.60-0.80), cross-check with medical keyword dictionary to adjust confidence

Layer 3 - Context Awareness: Consider employee history, temporal patterns, department risk profile

Layer 4 - Confidence-Based Routing:

Confidence Level	Action
≥0.85 (High)	Auto-route, no review needed
0.70-0.84 (Medium)	Auto-route with flag for verification
0.50-0.69 (Low)	Route to manual review queue
<0.50 (Very Low)	Default to NON-HEALTH

4.4 Routing Decision Matrix

Classification	Sub-Category	Destination Team	SLA
HEALTH	Emergency	Emergency Medical / Hospital	15 min
HEALTH	Mental Health	Mental Health Counselor	4 hours
HEALTH	Injury/Accident	Occupational Health Nurse	2 hours
HEALTH	Routine Medical	General Medical Team	24 hours
HEALTH	Insurance	HR Benefits Team	24 hours
NON-HEALTH	IT Support	IT Help Desk	4 hours
NON-HEALTH	HR Admin	HR Department	24 hours
UNCERTAIN	Low Confidence	Manual Review Queue	8 hours

4.5 Advantages of Cloud-Based LLM APIs Over Traditional NLP

Aspect	Traditional NLP	Cloud LLM APIs
Development Time	3-6 months	1-2 weeks
Labeled Data Required	5,000-50,000	0-50 examples
Annotation Cost	$10,000-$50,000	$0-$500
Initial Accuracy	60-70%	75-92%
Infrastructure Cost	$5,000-$20,000	Pay-per-use
Maintenance	High (retrain quarterly)	Low (prompt updates)
Scalability	Limited	Unlimited
Adaptation Time	Weeks	Hours

Multilingual	Separate models	Native support
Explainability	Limited	High (reasoning available)

Cost Comparison (Annual):

Cost Category	Traditional	LLM API
Year 1 Total	$105,000	$17,600
Year 2+	$55,000/year	$7,600/year
Savings	-	**$87,400 Year 1**

Key Advantages:

1. **No training data requirement** - Works with zero labeled examples
2. **Rapid deployment** - 2 weeks vs. 4 months
3. **70-80% cost reduction** - Lower upfront and ongoing costs
4. **Superior accuracy** - 85-92% vs. 60-70%
5. **Easy adaptation** - Update prompts in hours, not weeks
6. **No infrastructure management** - Provider handles scaling
7. **Built-in multilingual support** - Single prompt for 50+ languages

5. Implementation Recommendations

5.1 Technical Stack

- **LLM Provider:** GPT-4 or Claude 3.5 (primary), Gemini Pro (backup)
- **Integration:** RESTful API with retry logic
- **Caching:** Redis for duplicate classifications
- **Database:** PostgreSQL for metadata and logs
- **Monitoring:** Datadog/New Relic for latency tracking
- **Security:** API key rotation, PII redaction

5.2 Performance Metrics

Metric	Target
Classification Accuracy	≥85%
Precision (Health)	≥80%
Recall (Health)	≥85%
API Latency (P95)	<5 seconds
Routing Accuracy	≥90%

5.3 Continuous Improvement

- **Weekly:** Analyze misclassifications
- **Monthly:** Refine prompts based on errors
- **Quarterly:** Compare with human expert classification
- **Annual:** Review classification taxonomy

6. Conclusion

Large Language Models offer a compelling solution for classifying health-related tickets without labeled training data. Key benefits include:

1. **Immediate deployment** without data collection delays
2. **High accuracy** (85-92%) competitive with traditional models
3. **70-80% cost reduction** compared to traditional ML
4. **Rapid adaptation** with simple prompt updates
5. **Effortless scaling** using cloud infrastructure

Key Success Factors:

- Effective prompt engineering with clear class definitions
- Confidence-based routing with human-in-the-loop
- Continuous monitoring and feedback integration
- Multi-provider strategy for reliability

Future Enhancements:

- Fine-tuning specialized models on organizational data
- Semantic search for similar historical tickets
- Predictive urgency classification
- Integration with electronic health records